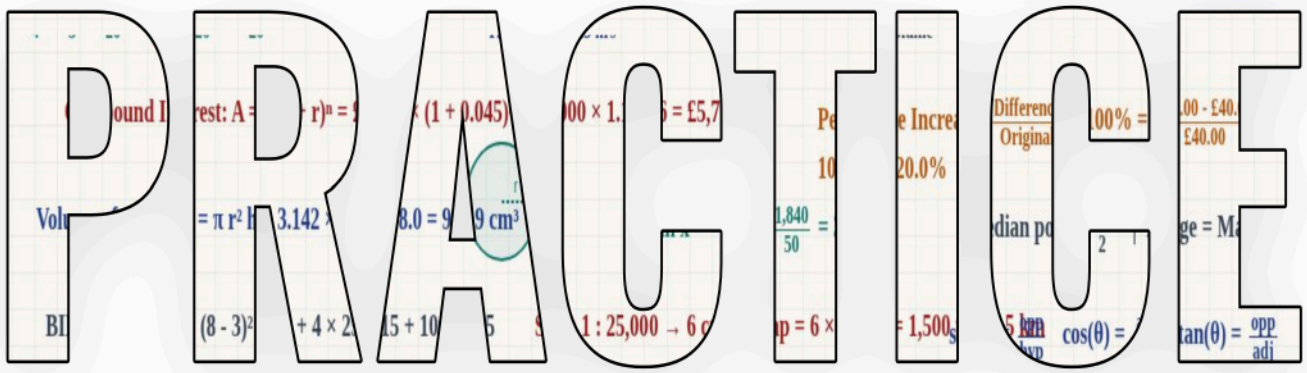
MASTER FUNCTIONAL SKILLS

FUNCTIONAL SKILLS MATHEMATICS

COMPANION PRACTICE WORKBOOK • STEPPED DRILLS

LEVEL 2



PRACTICE COMPANION

120+ stepped exam drills & full method mark workspaces

120+ Drills | Method Workspaces | Non-Calc & Calc | Worked Solutions

MATHS LEVEL 2

Functional Skills Mathematics

Level 2 Companion Workbook • 5mm Grid Candidate Working Frames & Full Mocks

This book belongs to:	Exam date:	Target level:
_____	_____	_____

What is inside

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How to use this book

Read
Read the rule. Study the picture.

Try
Write your answers on the lines.

Check
Mark it with the answers at the back.

Remember
Learn the Golden Rule by heart.

DRILL 1 OF 34 • LEVEL 2 STANDARDS

1 Worksheet 1: Negative Numbers & Overdrafts

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Rapid Warm-Up Drills:** Evaluate without a calculator: **(4 marks)**

a) -14 + (-9) = _____ b) 12 - (-18) = _____

c) (-6) × (-7) = _____ d) (-72) ÷ (+8) = _____

Show your working

Answer: _____

2 **Task 2: Workplace Context — Commercial Overdraft:** A transport firm has a bank overdraft balance of **-£2,340** on Monday morning. During the week: client payment +£5,800, diesel fuel -£1,950, insurance direct debit -£780, and maintenance -£1,150. Calculate the closing balance. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 1: Negative Numbers & Overdrafts • Candidate Workspace

✎ Show all working for Worksheet 1 tasks below. Check your final answers.

DRILL 2 OF 34 • LEVEL 2 STANDARDS

2

Worksheet 2: Column Multiplication & Decimals

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Column Multiplication Warm-Up:** Calculate without using a calculator. Show your column or grid workings clearly: **(4 marks)**

a) 348 × 27 b) 516 × 38

Show your working

Answer: _____

2 **Task 2: Bus Stop Division with Decimals:** Divide **918 by 24** using short or long division. Give your answer as an exact decimal. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 2: Column Multiplication & Decimals • Candidate Workspace

✎ Show all working for Worksheet 2 tasks below. Check your final answers.

DRILL 3 OF 34 • LEVEL 2 STANDARDS

3 Worksheet 3: Fraction Operations & Timber

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Fraction Operations Warm-Up:** Evaluate giving answers in simplest form: a) 5/12 + 3/8 b) 2 1/4 - 1 2/3. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Joinery Construction Problem:** A joiner cuts two battens: 1 3/4 m and 2 3/8 m from a 5 m timber plank. What length of plank remains? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 3: Fraction Operations & Timber • Candidate Workspace

✎ Show all working for Worksheet 3 tasks below. Check your final answers.

DRILL 4 OF 34 • LEVEL 2 STANDARDS

4 Worksheet 4: Fractions Multiply/Divide & Bakery

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Multiplication & Division Drills:** Evaluate giving answers in simplest form: a) 4/9 × 3/8 b) 3 1/2 ÷ 7/10. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Commercial Bakery Batching:** A baker uses 2 1/4 kg of flour per cake batch. He has 18 kg of flour in stock. How many batches can he bake? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 4: Fractions Multiply/Divide & Bakery • Candidate Workspace

✎ Show all working for Worksheet 4 tasks below. Check your final answers.

5

DRILL 5 OF 34 • LEVEL 2 STANDARDS

Worksheet 5: Decimals Arithmetic & Garage Invoice

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Decimal Arithmetic Warm-Up:** Calculate: a) 24.85 + 9.382 **(4 marks)**
b) 100 - 37.64 c) 4.6 × 0.35 d) 15.75 ÷ 0.5.

Show your working

Answer: _____

2 **Task 2: Mechanics Parts Invoice:** An auto garage invoice lists: 4 spark plugs at £14.25 each, brake pads at £68.50, and 2.5 hours labour at £42.00/hr. Find the pre-VAT total. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 5: Decimals Arithmetic & Garage Invoice • Candidate Workspace

✎ Show all working for Worksheet 5 tasks below. Check your final answers.

DRILL 6 OF 34 • LEVEL 2 STANDARDS

6 Worksheet 6: Percentages & Clearance Sale

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Multiplier Quick Fire:** State the single multiplier for: a) Increase by 17.5% b) Decrease by 12% c) Increase by 4.5% d) Find 8.25%. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Retail Clearance Discount:** A television originally priced at £650.00 is discounted by 22%. Calculate the new sale price. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 6: Percentages & Clearance Sale • Candidate Workspace

✎ Show all working for Worksheet 6 tasks below. Check your final answers.

DRILL 7 OF 34 • LEVEL 2 STANDARDS

7 Worksheet 7: Reverse Percentages & VAT

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Reverse Multiplier Warm-Up:** Find the original value: a) Value after 20% increase is £144 b) Value after 15% decrease is £204. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Pre-VAT Trade Price:** A tradesperson buys a cement mixer for £348.00 including 20% VAT. How much did the mixer cost before VAT was added? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 7: Reverse Percentages & VAT • Candidate Workspace

✎ Show all working for Worksheet 7 tasks below. Check your final answers.

DRILL 8 OF 34 • LEVEL 2 STANDARDS

8 Worksheet 8: Ratio Sharing & Business Profit

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Ratio Warm-Up Drills:** a) Share £560 in ratio 5 : 2. b) Express the ratio 15 cm : 1.2 m in simplest form 1 : n. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Partnership Profits:** Two directors share company profit in ratio 7 : 4. Director A receives £21,000 more than Director B. What was total profit? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 8: Ratio Sharing & Business Profit • Candidate Workspace

✎ Show all working for Worksheet 8 tasks below. Check your final answers.

DRILL 9 OF 34 • LEVEL 2 STANDARDS

9 Worksheet 9: Proportion & Unit Value

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Inverse Proportion Warm-Up:** a) 4 builders take 6 days to lay paving. How long for 3 builders at same pace? b) A car at 60 mph takes 4 hours. How long at 48 mph? **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Unit Pricing Best Buy:** Paint Brand X: 5 litres for £28.50. Paint Brand Y: 7.5 litres for £41.25. Which offers better value per litre? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 9: Proportion & Unit Value • Candidate Workspace

✎ Show all working for Worksheet 9 tasks below. Check your final answers.

DRILL 10 OF 34 • LEVEL 2 STANDARDS

10

Worksheet 10: Compound Interest & Van Depreciation

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Compound Multiplier Drills:** Calculate: a) £4,000 at 3% compound interest for 2 years. b) £12,000 depreciating at 10% for 3 years. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Fleet Van Depreciation:** A delivery van cost £24,000 brand new. It depreciates by 15% in year 1 and 12% in year 2. What is its value after 2 years? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 10: Compound Interest & Van Depreciation • Candidate Workspace

✎ Show all working for Worksheet 10 tasks below. Check your final answers.

DRILL 11 OF 34 • LEVEL 2 STANDARDS

11 Worksheet 11: BIDMAS & Braking Formula

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: BIDMAS & Powers Warm-Up:** Evaluate: a) $18 - 4 \times 3 + 5$ **(4 marks)**
b) $2 \times 3^2 - (10 - 4)$ c) $(45 - 3^2) \div (4 + 2)$ d) $5^2 - \sqrt{64}$.

Show your working

Answer: _____

2 **Task 2: Mechanics Braking Formula:** Stopping distance formula: **(3 marks)**
$d = 0.05v^2 + 0.2v$, where v = speed in mph. Find stopping distance at 60 mph.

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 11: BIDMAS & Braking Formula • Candidate Workspace

✎ Show all working for Worksheet 11 tasks below. Check your final answers.

DRILL 12 OF 34 • LEVEL 2 STANDARDS

12 Worksheet 12: Metric Conversions & Medicine

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Rapid Conversion Drills:** Convert: a) 4.8 kg into grams b) 650 ml into litres c) 2.4 m into cm d) 35 miles to km (5 mi ≈ 8 km). **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Hospital Patient Medication:** A patient requires 15 ml of oral antibiotic 3 times daily for 10 days. How many 200 ml bottles must be dispensed? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 12: Metric Conversions & Medicine • Candidate Workspace

✎ Show all working for Worksheet 12 tasks below. Check your final answers.

DRILL 13 OF 34 • LEVEL 2 STANDARDS

13

Worksheet 13: Elapsed Time & Meeting

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Elapsed Time Warm-Up:** Find elapsed duration: a) 08:45 to 14:15 b) 19:50 to 02:25 (next day). **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Train & Meeting Connection:** A business consultant catches the 09:28 train. Travel time is 1 hr 45 mins, followed by a 15-minute taxi ride. Does she arrive before 11:30? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 13: Elapsed Time & Meeting • Candidate Workspace

✎ Show all working for Worksheet 13 tasks below. Check your final answers.

DRILL 14 OF 34 • LEVEL 2 STANDARDS

14 Worksheet 14: Overtime & Income Tax

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Overtime Rate Calculations:** Standard rate £12.40/hr. **(4 marks)**
Find: a) Time-and-a-half rate b) Double time rate c) Pay for 35 normal hrs + 6 hrs at time-and-a-half.

Show your working

Answer: _____

2 **Task 2: Annual PAYE Income Tax:** A software developer earns £32,570 per annum. Personal allowance is £12,570. Basic rate tax is 20%. Calculate his annual income tax. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 14: Overtime & Income Tax • Candidate Workspace

✎ Show all working for Worksheet 14 tasks below. Check your final answers.

DRILL 15 OF 34 • LEVEL 2 STANDARDS

15 Worksheet 15: Perimeter & Skirting Board

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: L-Shape Perimeter Warm-Up:** An L-shaped room has overall length 9 m and overall width 7 m. Cutout corner is 4 m by 3 m. Find its perimeter. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Skirting Board Cost:** A rectangular bedroom measures 4.8 m by 3.6 m. It has a 0.9 m door opening. Skirting board costs £4.50 per metre. Calculate cost. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 15: Perimeter & Skirting Board • Candidate Workspace

✎ Show all working for Worksheet 15 tasks below. Check your final answers.

DRILL 16 OF 34 • LEVEL 2 STANDARDS

16

Worksheet 16: Circumference & Semicircular Bed

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Circumference Drills:** Calculate circumference (take π = 3.14): a) Circle with diameter 15 cm b) Circle with radius 6.2 m. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Semicircular Flowerbed Border:** A garden has a semicircular flowerbed with diameter 8 m. Brick edging is placed around the entire perimeter. Calculate total length. (Use π = 3.142) **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 16: Circumference & Semicircular Bed • Candidate Workspace

✎ Show all working for Worksheet 16 tasks below. Check your final answers.

17

DRILL 17 OF 34 • LEVEL 2 STANDARDS

Worksheet 17: Area & Laminate Flooring

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Area Formula Warm-Up:** Calculate area: a) Triangle: base 12 cm, height 7 cm b) Parallelogram: base 9.5 m, perpendicular height 4.2 m. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Laminate Flooring & 10% Wastage:** A dining room measures 5.4 m by 3.8 m. Flooring packs cover 2.2 m² each. Adding 10% for wastage, how many packs are needed? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 17: Area & Laminate Flooring • Candidate Workspace

✎ Show all working for Worksheet 17 tasks below. Check your final answers.

DRILL 18 OF 34 • LEVEL 2 STANDARDS

18 Worksheet 18: Trapezium & Pond Paving

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Trapezium Area Warm-Up:** Calculate area: a) Trapezium: parallel sides 6 m and 10 m, height 4 m b) Circle of radius 5 m (take π = 3.14). **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Paving Around a Circular Pond:** A square lawn of side 12 m has a circular pond of diameter 4 m in its centre. What is the area of grass remaining? (Use π = 3.142) **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 18: Trapezium & Pond Paving • Candidate Workspace

✎ Show all working for Worksheet 18 tasks below. Check your final answers.

DRILL 19 OF 34 • LEVEL 2 STANDARDS

19

Worksheet 19: Surface Area & Tent Fabric

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Cuboid Surface Area Warm-Up:** Calculate surface area: **(4 marks)**
a) Cuboid 10 cm × 6 cm × 4 cm b) Cube of side 5 cm.

Show your working

Answer: _____

2 **Task 2: Triangular Prism Tent Fabric:** A tent is a triangular prism. Triangular ends have base 2.4 m, height 1.6 m, and slant edges 2.0 m. Length is 3.5 m. Find total fabric area. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 19: Surface Area & Tent Fabric • Candidate Workspace

✎ Show all working for Worksheet 19 tasks below. Check your final answers.

DRILL 20 OF 34 • LEVEL 2 STANDARDS

20 Worksheet 20: Volume & Water Cistern

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Volume Calculation Warm-Up:** Calculate volume: a) Cuboid 8 cm × 5 cm × 12 cm b) Cylinder: radius 4 cm, height 10 cm (take π = 3.14). **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Water Storage Cistern:** A rectangular water tank is 1.8 m long, 1.2 m wide, and 0.8 m deep. How many litres of water does it hold when full? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 20: Volume & Water Cistern • Candidate Workspace

✎ Show all working for Worksheet 20 tasks below. Check your final answers.

DRILL 21 OF 34 • LEVEL 2 STANDARDS

21

Worksheet 21: Prism Density & Swimming Pool

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Prism Volume Drills:** A prism has cross-sectional area 45 cm² and length 18 cm. What is its volume? If it is made of aluminium with density 2.7 g/cm³, find its mass. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Sloping Swimming Pool Volume:** A pool is 15 m long and 6 m wide. Depth slopes uniformly from 1.0 m to 2.0 m. Calculate total volume of water in cubic metres. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 21: Prism Density & Swimming Pool • Candidate Workspace

✎ Show all working for Worksheet 21 tasks below. Check your final answers.

DRILL 22 OF 34 • LEVEL 2 STANDARDS

22

Worksheet 22: Scale Drawings & Floor Plan

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Scale Factor Warm-Up:** On a plan with scale 1 : 50: a) A wall is 6.5 cm on plan. Real length in metres? b) A sofa is 1.8 m long in reality. Plan length in cm? **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Office Suite Floor Plan:** On a 1 : 75 scale drawing, an office measures 12 cm by 8 cm. Find its real ground floor area in square metres. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 22: Scale Drawings & Floor Plan • Candidate Workspace

✎ Show all working for Worksheet 22 tasks below. Check your final answers.

DRILL 23 OF 34 • LEVEL 2 STANDARDS

23 Worksheet 23: Bearings & Ship Course

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Bearings Basics Warm-Up:** Write as 3-figure bearings: a) North-East b) South-West c) Due West d) 35° East of North. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Ship Course & Return Bearing:** A coastguard cutter sails from port on a bearing of 078° to reach a vessel in distress. What bearing must it steer to return to port? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 23: Bearings & Ship Course • Candidate Workspace

✎ Show all working for Worksheet 23 tasks below. Check your final answers.

DRILL 24 OF 34 • LEVEL 2 STANDARDS

24 Worksheet 24: Speed, Density & Courier Route

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Formula Calculations Warm-Up:** a) Speed: 105 miles in 1 hr 45 mins. b) Density: mass 840 g, volume 120 cm³. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Courier Van Delivery Route:** A delivery van travels 90 miles on a motorway at an average speed of 60 mph, then 15 miles through city traffic at 20 mph. Find total time. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 24: Speed, Density & Courier Route • Candidate Workspace

✎ Show all working for Worksheet 24 tasks below. Check your final answers.

DRILL 25 OF 34 • LEVEL 2 STANDARDS

25 Worksheet 25: Two-Way Tables & Transport

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Complete the Table:** 50 office workers surveyed on transport mode: 28 are Female. 12 Males drive. 18 Females take public transport. 6 Males walk. No Females walk. Complete the table. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Probability from Table:** From the table in Task 1, what fraction of Male workers drive? Give your answer in simplest form. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 25: Two-Way Tables & Transport • Candidate Workspace

✎ Show all working for Worksheet 25 tasks below. Check your final answers.

DRILL 26 OF 34 • LEVEL 2 STANDARDS

26 Worksheet 26: Dual Bar Charts & Sales Growth

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Read Dual Bar Chart:** A dual bar chart shows laptop sales (Units): Q1 (Store A: 120, Store B: 90); Q2 (Store A: 150, Store B: 180). Find total units sold and Store B's share. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Percentage Growth Comparison:** Store A sales grew from 120 to 150. Store B sales grew from 90 to 180. Calculate the percentage increase for both stores. Which grew faster? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 26: Dual Bar Charts & Sales Growth • Candidate Workspace

✎ Show all working for Worksheet 26 tasks below. Check your final answers.

DRILL 27 OF 34 • LEVEL 2 STANDARDS

27 Worksheet 27: Line Graphs & Energy Trend

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Temperature Monitoring:** Patient temperature logged: 08:00 (38.6°C), 10:00 (38.2°C), 12:00 (37.4°C). Estimate temperature at 09:00. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Energy Consumption Trend:** Electricity meter reading: Mon 420 kWh, Tue 460 kWh, Wed 510 kWh, Thu 480 kWh, Fri 430 kWh. Find mean daily usage. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 27: Line Graphs & Energy Trend • Candidate Workspace

✎ Show all working for Worksheet 27 tasks below. Check your final answers.

DRILL 28 OF 34 • LEVEL 2 STANDARDS

28 Worksheet 28: Pie Charts & Voter Angle

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Sector Angle Calculations:** 90 employees travel to work: Car (45), Bus (25), Train (15), Cycle (5). Calculate the sector angle for each travel mode. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Extracting Values from Angle:** In a pie chart of 1,200 voters, the sector for Candidate Green has an angle of 54°. How many people voted for Candidate Green? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 28: Pie Charts & Voter Angle • Candidate Workspace

✎ Show all working for Worksheet 28 tasks below. Check your final answers.

DRILL 29 OF 34 • LEVEL 2 STANDARDS

29 Worksheet 29: Averages & Adding Values

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Calculate the 4 Measures:** For data set: 14, 18, 12, 25, 18, 16, 23. Find: a) Mean b) Median c) Mode d) Range. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Adding a New Value to Mean:** A call centre agent has a mean call time of 4.5 minutes across 8 calls. Her 9th call lasts 9.0 minutes. What is her new mean? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 29: Averages & Adding Values • Candidate Workspace

✎ Show all working for Worksheet 29 tasks below. Check your final answers.

DRILL 30 OF 34 • LEVEL 2 STANDARDS

30 Worksheet 30: Grouped Frequency & Estimated Mean

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Midpoints & Products Warm-Up:** Classes: 0-10 (freq 4), 10-20 (freq 11), 20-30 (freq 5). State midpoints and calculate Sum(fx). **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Estimated Mean Calculation:** Using Task 1 above, calculate the estimated mean. Why is this an estimate? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 30: Grouped Frequency & Estimated Mean • Candidate Workspace

✎ Show all working for Worksheet 30 tasks below. Check your final answers.

DRILL 31 OF 34 • LEVEL 2 STANDARDS

31 Worksheet 31: Data Comparison & Drivers

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Delivery Driver Performance:** Driver X: Mean deliveries = 32, Range = 4. Driver Y: Mean deliveries = 36, Range = 15. Compare their performance. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Hospital Waiting Times:** Hospital A median wait = 42 mins, IQR = 15 mins. Hospital B median wait = 35 mins, IQR = 28 mins. Advise a patient. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 31: Data Comparison & Drivers • Candidate Workspace

✎ Show all working for Worksheet 31 tasks below. Check your final answers.

DRILL 32 OF 34 • LEVEL 2 STANDARDS

32

Worksheet 32: Scatter Graphs & Line of Best Fit

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Correlation Identification:** State correlation: a) Revision hours and exam score b) Car engine size and mpg c) Shoe size and salary. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Reading Line of Best Fit:** A scatter graph relates temperature (°C) to cold drinks sold. The line passes through (15, 60) and (25, 100). Predict drinks at 20°C. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 32: Scatter Graphs & Line of Best Fit • Candidate Workspace

✎ Show all working for Worksheet 32 tasks below. Check your final answers.

DRILL 33 OF 34 • LEVEL 2 STANDARDS

33 Worksheet 33: Probability Drills & Coin/Dice

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Basic Probability Drills:** A box has 8 red, 5 green, 7 blue counters. A counter is chosen at random: a) P(Red) b) P(Not Green) c) P(Yellow). **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Independent Combined Probability:** A fair coin is flipped and a standard 6-sided dice is rolled. What is the probability of getting a Head and a 6? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 33: Probability Drills & Coin/Dice • Candidate Workspace

✎ Show all working for Worksheet 33 tasks below. Check your final answers.

DRILL 34 OF 34 • LEVEL 2 STANDARDS

34 Worksheet 34: Combined Probability & Weather

★ Guided Calculation Scaffold

Step 1: Identify the statutory formula and extract key numerical variables from the text.
Step 2: Substitute values into calculation and record intermediate units.
Step 3: Perform secondary check using reverse arithmetic or approximation.

1 **Task 1: Complete Tree Branches:** Candidate sits two tests: P(Pass 1st) = 0.8. If pass, P(Pass 2nd) = 0.7. If fail, P(Pass 2nd) = 0.4. Find P(Pass both). **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Weather & Commute Delay:** P(Rain) = 0.3. If rain, P(Late) = 0.6. If dry, P(Late) = 0.1. Calculate total probability of being late. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 34: Combined Probability & Weather • Candidate Workspace

✎ Show all working for Worksheet 34 tasks below. Check your final answers.

CP1

CANDIDATE ASSESSMENT FRAME

Checkpoint Assessment 1 • Part 1/4

1 **Question 1:** Calculate without using a calculator: 416 × 28. Show your full column method. **(2 marks)**

Show your working

Answer: _____

2 **Question 2:** A freezer compartment is at -14°C. The thermostat is adjusted so the temperature rises by 6.5°C, then drops by 3.2°C. What is the new temperature? **(2 marks)**

Show your working

Answer: _____

3 **Question 3:** Evaluate giving your answer as a mixed number in simplest form: 1 3/5 + 2 1/4. **(2 marks)**

Show your working

Answer: _____

CP1

CANDIDATE ASSESSMENT FRAME

Checkpoint Assessment 1 • Part 2/4

3 **Question 4:** Work out: 3,744 ÷ 13. Show your full written division method. **(2 marks)**

Show your working

Answer: _____

4 **Question 5:** A tradesperson quotes £420.00 excluding VAT. VAT is charged at 20%. How much is the total quote with VAT included? **(2 marks)**

Show your working

Answer: _____

5 **Question 6:** Share £1,260 between three apprentices in the ratio 3 : 2 : 1. How much does the apprentice with the smallest share receive? **(2 marks)**

Show your working

Answer: _____

CP1

CANDIDATE ASSESSMENT FRAME

Checkpoint Assessment 1 • Part 3/4

5 **Question 7:** A sports retailer sells running shoes for £68.00 after a 15% discount. Calculate the original full price before the discount was applied. **(3 marks)**

Show your working

Answer: _____

6 **Question 8:** A company deposits £15,000 in a business growth account offering 3.8% compound interest per year for 3 years. Calculate the total balance in the account at the end of the 3 years. **(3 marks)**

Show your working

Answer: _____

7 **Question 9:** It takes 4 assembly technicians 15 hours to assemble a batch of medical ventilators. The hospital requests the batch in 10 hours. How many technicians in total are required? **(3 marks)**

Show your working

Answer: _____

CP1

CANDIDATE ASSESSMENT FRAME

Checkpoint Assessment 1 • Part 4/4

7 **Question 10:** A concrete mix requires sand, ballast, and cement in the ratio 5 : 3 : 2. A builder prepares 3.6 tonnes of mix. He has 1.7 tonnes of sand in stock. Does he have enough sand? Show full working. **(3 marks)**

Show your working

Answer: _____

8 **Question 11:** A logistics manager bought an electric delivery van for £32,000. It depreciates by 16% in the first year and 12% in the second year. What is the value of the van at the end of year 2? **(3 marks)**

Show your working

Answer: _____

9 **Question 12:** An office manager checks an invoice: 12 desks at £145.50 each, 24 chairs at £68.00 each. A delivery charge of £45.00 is added. A bulk discount of 7.5% is deducted from the total before adding 20% VAT. What is the final invoice total? **(3 marks)**

Show your working

Answer: _____

CANDIDATE ASSESSMENT FRAME

CP2

Checkpoint Assessment 2 • Part 1/4

1 **Question 1:** A delivery truck travels 135 miles in 2 hours 15 minutes. Calculate its average speed in miles per hour. **(2 marks)**

Show your working

Answer: _____

2 **Question 2:** Convert: a) 4.2 kg to grams b) 650 ml to litres c) 25 miles to km using 5 miles = 8 km. **(2 marks)**

Show your working

Answer: _____

3 **Question 3:** A circle has a diameter of 14 cm. Using π = 22/7, calculate: a) its circumference b) its area. **(2 marks)**

Show your working

Answer: _____

CANDIDATE ASSESSMENT FRAME

CP2

Checkpoint Assessment 2 • Part 2/4

3 **Question 4:** A rectangular courtyard measures 12 m by 8.5 m. In the centre is a square fountain with sides of 2.5 m. Calculate the paved area of the courtyard. **(2 marks)**

Show your working

Answer: _____

4 **Question 5:** A clock shows 09:35. An online training conference lasts for 3 hours 48 minutes. At what time does the conference end? (Use 24-hour clock format) **(2 marks)**

Show your working

Answer: _____

5 **Question 6:** A warehouse worker earns £12.50 per hour for 36 hours. Overtime is paid at time-and-a-half. In one week he works 4 hours overtime. What is his gross pay? **(2 marks)**

Show your working

Answer: _____

CP2

CANDIDATE ASSESSMENT FRAME

Checkpoint Assessment 2 • Part 3/4

5 **Question 7:** A commercial grain silo is shaped as a cylinder with diameter 4.2 m and height 8.5 m. Calculate the volume of the silo in cubic metres to 1 decimal place. (Use π = 3.142) **(3 marks)**

Show your working

Answer: _____

6 **Question 8:** A metal casting has a mass of 3,570 g and a volume of 420 cm³. Calculate its density in g/cm³. If pure zinc has a density of 7.14 g/cm³, determine whether the casting is pure zinc. **(3 marks)**

Show your working

Answer: _____

7 **Question 9:** On a map with a scale of 1 : 25,000, the distance between two telecommunication masts is 16.4 cm. Calculate the actual distance between the masts in kilometres. **(3 marks)**

Show your working

Answer: _____

CANDIDATE ASSESSMENT FRAME

CP2

Checkpoint Assessment 2 • Part 4/4

7 **Question 10:** An office worker earns £34,000 gross per year. The tax-free personal allowance is £12,570. Basic rate income tax is charged at 20%. Employee National Insurance is 8% on earnings above £12,570. Calculate her total monthly take-home pay after tax and NI. **(3 marks)**

Show your working

Answer: _____

8 **Question 11:** A lawn is in the shape of a trapezium with parallel sides 14 m and 22 m, and perpendicular distance 12 m. Grass seed costs £6.40 per box. Each box covers 25 m². How much does it cost to seed the entire lawn? **(3 marks)**

Show your working

Answer: _____

9 **Question 12:** A heating engineer installs a closed cylindrical thermal expansion vessel: radius 0.35 m and height 1.2 m. The entire exterior surface is insulated with thermal foil costing £14.50/m². Calculate the cost of the foil. (Use π = 3.142) **(3 marks)**

Show your working

Answer: _____

CP3

CANDIDATE ASSESSMENT FRAME

Checkpoint Assessment 3 • Part 1/4

1 **Question 1:** The table shows the number of patients seen at a dental practice: Mon 24, Tue 32, Wed 28, Thu 36, Fri 30. Calculate: a) the mean b) the range. **(2 marks)**

Show your working

Answer: _____

2 **Question 2:** A spinner has 8 equal sections numbered 1 to 8. Calculate the probability of spinning: a) an even number b) a multiple of 3. **(2 marks)**

Show your working

Answer: _____

3 **Question 3:** 60 office workers were surveyed on their preferred lunch option: Sandwiches (30), Hot Food (20), Salad (10). Calculate the pie chart sector angle for each option. **(2 marks)**

Show your working

Answer: _____

CANDIDATE ASSESSMENT FRAME

CP3

Checkpoint Assessment 3 • Part 2/4

3 **Question 4:** A box contains 5 red pens, 7 blue pens, and 8 black pens. A pen is picked at random. What is the probability that it is NOT red? Give your answer as a fraction in simplest form. **(2 marks)**

Show your working

Answer: _____

4 **Question 5:** The test marks of 9 apprentices are: 45, 52, 58, 61, 64, 68, 72, 75, 85. Find: a) the median mark b) the interquartile range. **(2 marks)**

Show your working

Answer: _____

5 **Question 6:** State whether each pair of variables has positive, negative, or no correlation: a) Daily sunshine hours and solar panel output b) Car engine mileage and market resale value. **(2 marks)**

Show your working

Answer: _____

CANDIDATE ASSESSMENT FRAME

CP3

Checkpoint Assessment 3 • Part 3/4

5 **Question 7:** A delivery depot records the delivery times (t mins) for 50 consignments: $0 < t \leq 10$ (12), $10 < t \leq 20$ (22), $20 < t \leq 30$ (11), $30 < t \leq 40$ (5). Calculate the estimated mean delivery time. **(3 marks)**

Show your working

Answer: _____

6 **Question 8:** A quality assurance inspector tests computer microchips. Machine A produces 60% of chips with a 2% defect rate. Machine B produces 40% of chips with a 5% defect rate. What is the overall probability that a randomly chosen chip is defective? **(3 marks)**

Show your working

Answer: _____

7 **Question 9:** A scatter graph plots vehicle weight (kg) against fuel economy (mpg). The line of best fit passes through (1200, 48) and (1800, 32). Predict the fuel economy for a vehicle weighing 1500 kg, and describe the relationship. **(3 marks)**

Show your working

Answer: _____

CANDIDATE ASSESSMENT FRAME

CP3

Checkpoint Assessment 3 • Part 4/4

7 **Question 10:** A supermarket manager audits 200 customer transactions. 120 customers paid by contactless card, 50 by mobile app, and 30 in cash. Of the contactless users, 40% spent over £30. Of the mobile app users, 60% spent over £30. Construct a two-way table and find the probability that a customer picked at random spent over £30. **(3 marks)**

Show your working

Answer: _____

8 **Question 11:** Two recruitment consultants interview candidates. Consultant X rates 8 candidates with a mean score of 72. Consultant Y rates 12 candidates with a mean score of 65. Calculate the combined mean score for all 20 candidates. **(3 marks)**

Show your working

Answer: _____

9 **Question 12:** The probability of train service A being delayed is 0.20. The probability of train service B being delayed is 0.15. Assuming delays are independent, calculate the probability that at least one train is on time. **(3 marks)**

Show your working

Answer: _____

CS1

SYNOPTIC REAL-WORLD PROBLEM

Case Study 1: NHS Facilities & Ward Refurbishment

Project Specification

Calculate flooring area, antimicrobial paint requirements, and project budget compliance.

Use the data provided to complete the project costing and compliance audit.

1 **Task 1 (Flooring & Paint Costing):** A rectangular hospital ward measures 15 m long by 8 m wide, with walls 3 m high. Anti-slip vinyl flooring costs £24.50 per m². Antimicrobial wall paint covers 12 m² per tin and each tin costs £38.00 (ignore doors/windows for safety margin). Calculate the combined material cost for the floor and 2 coats of wall paint. **(4 marks)**

Show your working

Answer: _____

2 **Task 2 (Labour & Budget Compliance):** Three Estates operatives working 8 hours per day can complete the refurbishment in 5 days at £18.50 per hour each. If the Ward Manager has a total refurbishment budget of £5,200 (materials + labour), determine whether the project is within budget and state the surplus or deficit. **(4 marks)**

Show your working

Answer: _____

PROJECT CALCULATIONS FRAME

Case Study 1 • Candidate Workspace

✎ **Candidate full calculation working and decision summary:**

SYNOPTIC REAL-WORLD PROBLEM

CS2

Case Study 2: Logistics Hub & Pallet Loading Optimization

Project Specification

Determine volume capacities, weight axle tolerances, and delivery fuel efficiency.

Use the data provided to complete the project costing and compliance audit.

1 **Task 1 (Container Volume & Payload Limit):** An articulated trailer has internal dimensions 13.2 m long, 2.4 m wide, and 2.6 m high. Standard Euro-pallets measure 1.2 m by 0.8 m and are stacked to a height of 1.3 m (allowing 2 tiers high). Each loaded pallet weighs 380 kg. Calculate the maximum number of pallets that fit in the trailer and check if their total weight is within the 26-tonne axle payload limit. **(4 marks)**

Show your working

Answer: _____

2 **Task 2 (Fleet Fuel & Delivery Schedule):** The loaded lorry travels 210 miles at an average speed of 50 mph, plus a 45-minute statutory tachograph break. Fuel consumption is 8.4 miles per gallon (1 gallon = 4.55 litres) and diesel costs £1.48 per litre. Calculate the total journey time and total fuel cost. **(4 marks)**

Show your working

Answer: _____

PROJECT CALCULATIONS FRAME

Case Study 2 • Candidate Workspace

✎ **Candidate full calculation working and decision summary:**

SYNOPTIC REAL-WORLD PROBLEM

CS3

Case Study 3: Commercial Construction Site & Concrete Pour

Project Specification

Calculate trench volumes, foundation mix ratios, and site labour scheduling.

Use the data provided to complete the project costing and compliance audit.

1 **Task 1 (Trench Volume & Mix Ratio):** A foundation trench is 25 m long, 0.8 m wide, and 1.2 m deep. Concrete has a density of 2,400 kg/m³ and is mixed using cement, sand, and aggregate in the ratio 1 : 2 : 3 by mass. Calculate the trench volume in m³ and the mass of cement required in tonnes. **(4 marks)**

Show your working

Answer: _____

2 **Task 2 (Site Labour Inverse Proportion & Cost):** Six groundworkers take 8 working days to excavate and pour the foundations. To beat forecasted frost, the site manager wants the work completed in 4 working days. How many additional groundworkers must be hired, and what is the total labour bill if each worker is paid £165 per day? **(4 marks)**

Show your working

Answer: _____

PROJECT CALCULATIONS FRAME

Case Study 3 • Candidate Workspace

✎ **Candidate full calculation working and decision summary:**

P1

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 1 • Page 1 of 8

1 **Question 1:** Calculate without using a calculator: 648 × 35. Show your full written method. **(3 marks)**

Show your working

Answer: _____

2 **Question 2:** Evaluate giving your answer as a mixed number in simplest form: 2 2/3 ÷ 4/9. **(2 marks)**

Show your working

Answer: _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 1 • Page 2 of 8

3 **Question 3:** A kitchen fitter cuts a circular hole with diameter 28 cm in a wooden worktop. Using π = 22/7, calculate the perimeter (circumference) of the cut hole. **(3 marks)**

Show your working

Answer: _____

4 **Question 4:** An invoice shows a power saw sold for £264.00 including 20% VAT. Calculate the cost of the saw before VAT was added. **(3 marks)**

Show your working

Answer: _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 1 • Page 3 of 8

5 **Question 5:** A healthcare assistant prepares 750 ml of saline infusion. Each dose is 60 ml. How many full doses can be administered, and how many millilitres remain? **(2 marks)**

Show your working

Answer: _____

6 **Question 6:** Share £2,400 between three apprentices in the ratio 4 : 3 : 1. What is the difference between the largest share and the smallest share? **(3 marks)**

Show your working

Answer: _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 1 • Page 4 of 8

7 **Question 7:** A senior nurse has a gross salary of £36,570. The personal allowance is £12,570. Basic rate income tax is 20%. National Insurance is 8% of earnings above £12,570. A workplace pension deduction of 6% of gross salary is made. Calculate her net monthly take-home pay. **(4 marks)**

Show your working

Answer: _____

8 **Question 8:** An engineering firm purchases a computerized CNC milling machine for £45,000. The machine depreciates at a compound rate of 15% per annum. Calculate the book value of the machine after 3 years. **(4 marks)**

Show your working

Answer: _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 1 • Page 5 of 8

9 **Question 9:** A heavy haulage transporter travels 168 miles. The driver leaves depot at 06:15 and takes a mandatory 45-minute rest break during the journey. The vehicle arrives at 10:30. Calculate the driver's average driving speed in mph while moving. **(4 marks)**

Show your working

Answer: _____

10 **Question 10:** A landscape contractor constructs a patio in the shape of a trapezium with parallel sides 8.5 m and 13.5 m, and perpendicular height 6.4 m. Paving slabs cost £28.50 per square metre. Sand and mortar bedding adds £4.50 per square metre. Calculate the total material cost. **(4 marks)**

Show your working

Answer: _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 1 • Page 6 of 8

11 **Question 11:** A commercial stainless-steel cylindrical storage tank has an internal diameter of 2.4 m and a height of 3.5 m. Calculate the liquid capacity of the tank in litres. (Use π = 3.142) **(4 marks)**

Show your working

Answer: _____

12 **Question 12:** A local council tracks response times (hours) for emergency housing repairs: $0 < h \leq 12$ (18 calls), $12 < h \leq 24$ (32 calls), $24 < h \leq 36$ (14 calls), $36 < h \leq 48$ (6 calls). Calculate the estimated mean response time in hours. **(4 marks)**

Show your working

Answer: _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 1 • Page 7 of 8

13 **Question 13:** An architect designs an executive office suite: 12 m long by 8.5 m wide. A circular glass conference booth of diameter 4 m is located in the corner. The remaining floor space is carpeted with luxury tiles costing £32.00/m². VAT at 20% is added to the carpet cost. Calculate the total cost to carpet the office suite. (Use π = 3.142) **(5 marks)**

Show your working

Answer: _____

14 **Question 14:** On an ordnance survey map drawn to a scale of 1 : 25,000, a wildlife reserve boundary is a rectangle measuring 14.8 cm by 9.2 cm. Calculate the real perimeter of the reserve in kilometres, and the real land area of the reserve in square kilometres. **(5 marks)**

Show your working

Answer: _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 1 • Page 8 of 8

15 **Question 15:** A healthcare audit tracks 150 clinic patients tested for diabetes: 90 patients are over 50 years old. Of those over 50, 27 tested positive. Of those aged 50 or under, 12 tested positive. Construct a complete two-way table. A patient is chosen at random from those who tested positive. What is the probability that this patient is aged 50 or under? **(5 marks)**

Show your working

Answer: _____

16 **Question 16:** A manufacturing plant produces precision steel rods. A sample of 100 rods has a mean length of 148.5 mm with a range of 4.2 mm. A second production line produces 100 rods with a mean length of 148.4 mm with a range of 1.1 mm. The target length is 148.5 mm. Compare the performance of the two production lines and advise the plant manager which line produces higher quality output. **(5 marks)**

Show your working

Answer: _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 2 • Page 1 of 8

1 **Question 1:** Calculate: 684 ÷ 18 without using a calculator. Show your full working. **(4 marks)**

Show your working

Answer: _____

2 **Question 2:** A pallet measures 1.2 m by 0.8 m. A warehouse floor is 24 m by 16 m. Calculate the maximum number of pallets that can fit on the floor in a single layer. **(4 marks)**

Show your working

Answer: _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 2 • Page 2 of 8

3 **Question 3:** Calculate: 3/5 × 4/7. Express your answer in simplest form. **(4 marks)**

Show your working

Answer: _____

4 **Question 4:** A delivery driver's hourly rate increased from £12.50 to £13.75. Calculate the percentage pay increase. Check your answer. **(4 marks)**

Show your working

Answer: _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 2 • Page 3 of 8

5 **Question 1:** Calculate: 345.8 − 187.94 without a calculator. **(4 marks)**
Show columnar working.

Show your working

Answer: _____

6 **Question 2:** A retailer buys stock for £450 and sells it for £630. **(4 marks)**
Calculate the percentage profit margin on cost.

Show your working

Answer: _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 2 • Page 4 of 8

7 **Question 5: Fleet Fuel Efficiency:** Van A travels 420 miles on 60 litres of fuel. Van B travels 385 miles on 52 litres. Which van achieves better miles-per-gallon (mpg)? (1 gallon = 4.55 litres). **(5 marks)**

Show your working

Answer: _____

8 **Question 6: Cargo Weight & Density:** A container has volume 36 m^3. It is loaded with cargo having an average density of 650 kg/m^3. The maximum payload limit is 24 tonnes. Is the container overloaded? **(5 marks)**

Show your working

Answer: _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 2 • Page 5 of 8

9 **Question 7: Courier Delivery Routes:** Driver covers 168 miles in 3 hours 30 mins. A 45-minute delivery stop was included. What was the driving speed while moving? **(5 marks)**

Show your working

Answer: _____

10 **Question 8: Forklift Ramp Slope:** A loading ramp rises 1.2 m over a horizontal distance of 5 m. Calculate the length of the sloping surface using Pythagoras ($a^2 + b^2 = c^2$). **(5 marks)**

Show your working

Answer: _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 2 • Page 6 of 8

11 **Question 9: Delivery Time Statistics:** Delivery times (mins): 10-19 (5), 20-29 (14), 30-39 (21), 40-49 (10). Calculate the estimated mean delivery time. **(5 marks)**

Show your working

Answer: _____

12 **Question 10: Warehouse Racking:** Racking is offered with a 12% discount. The discounted cost is £3,960. Find the original price before discount. **(5 marks)**

Show your working

Answer: _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 2 • Page 7 of 8

13 **Question 11: Currency & Import:** A consignment from Germany costs €14,500. Customs import duty is 6%. If £1 = €1.15, calculate the total cost in GBP including duty. **(5 marks)**

Show your working

Answer: _____

14 **Question 12: Delivery Success Rate:** Out of 850 deliveries, 34 were failed deliveries. Calculate the percentage success rate to 1 decimal place. **(5 marks)**

Show your working

Answer: _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 2 • Page 8 of 8

15 **Question 13: Storage Space Volume:** A cold storage room is 12m long, 8m wide, and 3.5m high. Industrial cooling requires 45 Watts per m^3. What power cooling unit is required? **(4 marks)**

Show your working

Answer: _____

16 **Question 14: Van Tyre Replacement:** Tyres cost £95 each. A fleet replaces all 4 tyres on 15 vans. The supplier offers 'Buy 3 get 1 free' on every van. Calculate total cost. **(4 marks)**

Show your working

Answer: _____

MODEL ANSWERS & MARKING CRITERIA

Answer Key • Part 1 of 3

▶ Worksheets 1–12 Step-by-Step Model Answers

W1: T1: a) -23; b) 30; c) 42; d) -9. **T2:** Closing balance = -2340 + 5800 - 1950 - 780 - 1150 = **-£420.00** (in overdraft).

W2: T1: a) 348 × 27 = **9,396**; b) 516 × 38 = **19,608**. **T2:** 918 ÷ 24 = **38.25**.

W3: T1: a) (10+9) ÷ 24 = **19/24**; b) 9/4 - 5/3 = (27-20) ÷ 12 = **7/12**. **T2:** Cut = 1 3/4 + 2 3/8 = 4 1/8 m. Remaining = 5 - 4 1/8 = **7/8 m** (0.875 m).

W4: T1: a) 12/72 = **1/6**; b) 7/2 × 10/7 = **5**. **T2:** Batches = 18 ÷ 2 1/4 = 18 × 4/9 = **8 batches**.

W5: T1: a) 34.232; b) 62.36; c) 1.61; d) 31.5. **T2:** Pre-VAT = (4 × 14.25) + 68.50 + (2.5 × 42) = 57 + 68.50 + 105 = **£230.50**.

W6: T1: a) 1.175; b) 0.88; c) 1.045; d) 0.0825. **T2:** Discount = £650 × 0.22 = £143. Sale price = £650 × 0.78 = **£507.00**.

W7: T1: a) 144 ÷ 1.20 = **£120.00**; b) 204 ÷ 0.85 = **£240.00**. **T2:** Pre-VAT = £348 ÷ 1.20 = **£290.00**.

W8: T1: a) 560 ÷ 7 = 80 → **£400 and £160**; b) 15 cm : 120 cm = **1 : 8**. **T2:** Difference = 3 parts = £21,000 → 1 part = £7,000. Total profit (11 parts) = **£77,000**.

W9: T1: a) 4 × 6 = 24 → 24 ÷ 3 = **8 days**; b) 240 ÷ 48 = **5 hours**. **T2:** Brand X = 28.50 / 5 = £5.70/L; Brand Y = 41.25 / 7.5 = £5.50/L. **Brand Y is better value**.

W10: T1: a) 4000 × $(1.03)^2$ = **£4,243.60**; b) 12000 × $(0.9)^3$ = **£8,748.00**. **T2:** £24,000 × 0.85 × 0.88 = **£17,952.00**.

W11: T1: a) 11; b) 12; c) 6; d) 17. **T2:** d = 0.05(60^2) + 0.2(60) = 0.05(3600) + 12 = 180 + 12 = **192 feet**.

W12: T1: a) 4,800 g; b) 0.65 L; c) 240 cm; d) 56 km. **T2:** Total dose = 15 × 3 × 10 = 450 ml. Bottles needed = 450 ÷ 200 = 2.25 → **3 bottles**.

✎ Personal Learning Reflections (Part 1)

MODEL ANSWERS & MARKING CRITERIA

Answer Key • Part 2 of 3

▶ Worksheets 13–24 Step-by-Step Model Answers

W13: T1: a) 5h 30m; b) 6h 35m. **T2:** 09:28 + 1h 45m = 11:13 + 15m = 11:28. **Yes, arrives before 11:30** (at 11:28).

W14: T1: a) £18.60; b) £24.80; c) (35 × 12.40) + (6 × 18.60) = 434 + 111.60 = **£545.60**. **T2:** Taxable = £20,000. Tax (20%) = **£4,000.00**.

W15: T1: P = 2(9 + 7) = **32 m**. **T2:** Perimeter = 2(4.8 + 3.6) - 0.9 = 16.8 - 0.9 = 15.9 m. Cost = 15.9 × 4.50 = **£71.55**.

W16: T1: a) 3.14 × 15 = **47.1 cm**; b) 2 × 3.14 × 6.2 = **38.94 m**. **T2:** Curved = 1/2 π d = 1/2(3.142)(8) = 12.57 m. Total P = 12.57 + 8 = **20.57 m**.

W17: T1: a) 1/2(12)(7) = **42 cm²**; b) 9.5 × 4.2 = **39.9 m²**. **T2:** Area = 5.4 × 3.8 = 20.52 m². With 10% = 22.572 m². Packs = 22.572 ÷ 2.2 = 10.26 → **11 packs**.

W18: T1: a) (6+10) ÷ 2 × 4 = **32 m²**; b) 3.14 × 25 = **78.5 m²**. **T2:** Lawn = 144 m². Pond = 3.142 × 4 = 12.57 m². Remaining grass = **131.43 m²**.

W19: T1: a) 2(60 + 40 + 24) = **248 cm²**; b) 6 × 25 = **150 cm²**. **T2:** Ends = 2(1/2 × 2.4 × 1.6) = 3.84 m². Sides = 2(2.0 × 3.5) = 14 m². Base = 2.4 × 3.5 = 8.4 m². Total = **26.24 m²**.

W20: T1: a) 8 × 5 × 12 = **480 cm³**; b) 3.14 × 16 × 10 = **502.4 cm³**. **T2:** V = 1.8 × 1.2 × 0.8 = 1.728 m³ = **1,728 litres**.

W21: T1: V = 45 × 18 = 810 cm³. Mass = 810 × 2.7 = **2,187 g** (2.187 kg). **T2:** Mean depth = 1.5 m. V = 15 × 6 × 1.5 = **135 m³**.

W22: T1: a) 6.5 × 50 = 325 cm = **3.25 m**; b) 180 ÷ 50 = **3.6 cm**. **T2:** Real l = 12 × 0.75 = 9 m; w = 8 × 0.75 = 6 m. Area = 9 × 6 = **54 m²**.

W23: T1: a) 045°; b) 225°; c) 270°; d) 035°. **T2:** Return bearing = 078° + 180° = **258°**.

W24: T1: a) 105 ÷ 1.75 = **60 mph**; b) 840 ÷ 120 = **7.0 g/cm³**. **T2:** Motorway = 90/60 = 1.5 hrs; City = 15/20 = 0.75 hrs. Total = 2.25 hrs = **2 hours 15 minutes**.

✎ Personal Learning Reflections (Part 2)

MODEL ANSWERS & MARKING CRITERIA

Answer Key • Part 3 of 3

Worksheets 25–34 & Case Studies 1–3 Model Answers

W25: T1: Male: Drive 12, Public 4, Walk 6, Total 22. Female: Drive 10, Public 18, Walk 0, Total 28. **T2:** Fraction of males driving = 12/22 = **6/11**.

W26: T1: Total units = (120+90) + (150+180) = 540. Store B share = 270/540 = **50%**. **T2:** Store A = (30/120) × 100 = 25%; Store B = (90/90) × 100 = 100%. **Store B grew faster**.

W27: T1: Estimate at 09:00: mid-point between 38.6 and 38.2 = **38.4°C**. **T2:** Mean daily usage = (420+460+510+480+430) ÷ 5 = 2300 ÷ 5 = **460 kWh**.

W28: T1: Car = 180°, Bus = 100°, Train = 60°, Cycle = 20°. **T2:** Votes = (54/360) × 1,200 = **180 votes**.

W29: T1: a) Mean = 126/7 = **18**; b) Median = **18**; c) Mode = **18**; d) Range = 25-12 = **13**. **T2:** Total sum = (8 × 4.5) + 9.0 = 36 + 9 = 45. New mean = 45 ÷ 9 = **5.0 minutes**.

W30: T1: Midpoints: 5, 15, 25. Products: 20, 165, 125. Sum(fx) = 310. **T2:** Estimated mean = 310 ÷ 20 = **15.5**. It is an estimate because exact values within intervals are unknown.

W31: T1: Driver Y has higher mean deliveries (36 vs 32), but Driver X is much more consistent with a smaller range (4 vs 15). **T2:** Hospital B has lower median wait (35 vs 42 mins), but Hospital A has more consistent waits (IQR 15 vs 28 mins).

W32: T1: a) Positive; b) Negative; c) Zero. **T2:** At 20°C, drinks sold = **80 units** (midpoint between 60 and 100).

W33: T1: Total = 20. a) P(Red) = 8/20 = **2/5**; b) P(Not Green) = 15/20 = **3/4**; c) P(Yellow) = **0**. **T2:** P(Head and 6) = 1/2 × 1/6 = **1/12**.

W34: T1: P(Pass both) = 0.8 × 0.7 = **0.56** (56%). **T2:** P(Rain and Late) = 0.3 × 0.6 = 0.18. P(Dry and Late) = 0.7 × 0.1 = 0.07. Total P(Late) = 0.18 + 0.07 = **0.25** (25%).

CS1 (NHS Ward): Floor 120 m² (£2,940) + Paint 276 m² ÷ 12 = 23 tins (£874) = **£3,814 materials**. Labour 120 hrs × £18.50 = £2,220. Total = **£6,034 (£834 over budget)**.

CS2 (Logistics Hub): 11 × 3 × 2 = **66 pallets**; Mass = **25.08 t ≤ 26 t (OK)**. Time = **4h 57m**; Fuel = 25 gal = 113.75 L × £1.48 = **£168.35**.

CS3 (Concrete Pour): Volume = **24 m³**; Mass = 57.6 t → Cement (1/6) = **9.6 tonnes**. Workers for 4 days = 12 (**6 extra workers**); Labour = **£7,920.00**.

✎ Personal Learning Reflections (Part 3)

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978-1-912345-81-4